RECEIVING STATION OH2GAX · NEAR HELSINKI-VANTAA

# MODE-S Wind

## Airliners as weather sensors

A passive 1090 MHz receiving station that turns the replies of passing airliners into **wind, temperature, windshear and turbulence** — measured by the aircraft themselves, live, from the final approach up to cruise level.

Passive · listen only | Live every few seconds | Web + phone | Open source · GPLv3




**Otso Laakso · OH2GAX** · Version v.06112026 · November 2026 · 1 / 2

**≈ 130 /s** — Mode S and ADS-B messages heard, 24-hour mean

**160 NM** — range of the receiver at cruise levels

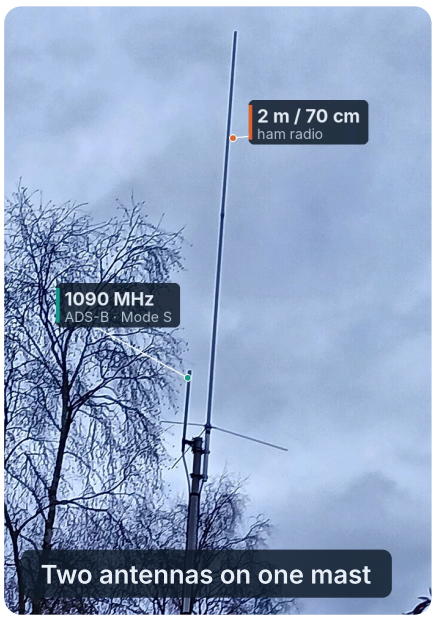
**≈ 200 + 200** — landings and departures followed every day

**0.5 s** — a headwind value of each aircraft on final and climb-out

**4.4 kt** — wind against the MEPS model, vector RMS over 21 days

**2 receivers** — merged in the cloud — and two servers, local and cloud

## 1 HOW IT WORKS — one antenna, one receiver, one small computer, a second receiver since Nov 2026; the users are served from the cloud










**Aircraft** 1
Radars ask on 1030 MHz; the transponders answer on 1090 MHz with airspeed, heading, Mach, track and ground speed (Mode S EHS) and broadcast GNSS position and velocity (ADS-B).

**Receiver** 2
Listens only: the Radarcape stamps every reply by GPS (MLAT); a second receiver adds more.

**Local station** 3
A Raspberry Pi 4 decodes, checks, calibrates and stores — also without the internet.

**Cloud server** 4
A full second copy at AWS, fed by outbound encrypted tunnels, serves the users over HTTPS.

**You** 5
any browser and the phone view **/app** — the address on page 2

## 2 WHAT THE AIRCRAFT MEASURE — the same numbers air traffic control uses, turned into weather

### Wind

$$\text{wind} = (\text{GS}, \text{track}) - (\text{TAS}, \text{heading})$$

The ground-speed vector minus the air vector, from two replies at most 10 s apart; the magnetic heading turned to true with the World Magnetic Model.

### Temperature

$$T = 288.15\ \text{K} \cdot (\text{TAS} / (a_0 \cdot M))^2$$

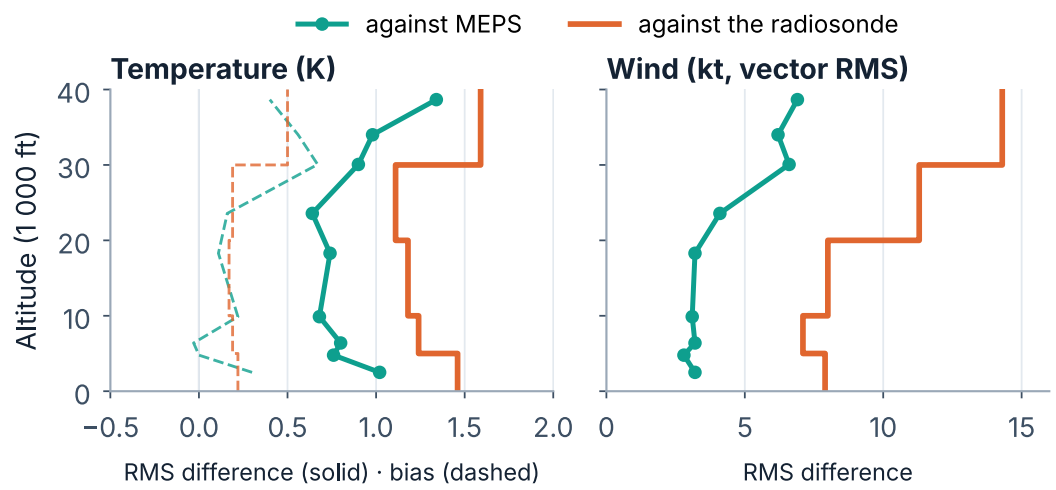
From the true airspeed and Mach number ($a_0$ = 661.47 kt); each reply is coarse, so profiles use trimmed means of many aircraft.

### Heights of the pressure surfaces

$$z = \text{pressure altitude} + \Delta_{\text{GNSS–baro}}$$

The pressure altitude says which surface the aircraft flies on, its ADS-B GNSS difference how high it lies: height contours and the geostrophic wind.

## 3 SELF-CORRECTING, INDEPENDENTLY CHECKED — against the radiosonde and the MEPS weather model




**How well the aircraft agree** — the system's own checks to 1 Nov 2026: the Mode S area profile against **MEPS** hourly (21 days, 298 hours) and the **Jokioinen radiosonde** (66 launches since 10 Oct); raw temperatures. The sonde flies ~100 km away: part of its difference is distance.

**0.9 K · 4.4 kt** — temperature · wind against MEPS (RMS, all levels)

**1.3 K · 9.7 kt** — against the radiosonde (RMS, 66 launches)

**11.5 → 9.1 kt**
**Heading offset of each airframe** — old magnetic-variation tables in the avionics, learnt per aircraft: wind against the soundings when introduced

**1.27 → 0.96 K**
**Upper-level temperature** — the fleet reads warm at 300–250 hPa; a correction learnt every hour: error against the soundings at 200–300 hPa

**± 2 m**
**GNSS heights** of the pressure surfaces against MEPS at 400–250 hPa, from altimeter and geoid corrections learnt at every sounding

© 2026 Otso Laakso (OH2GAX) · MODE-S Wind, GPLv3 · github.com/oh2gax/mode_s_wind

Decoding with pyModeS (Junzi Sun, TU Delft) · method after Siebren de Haan (KNMI)

## 4 LOW-LEVEL WINDSHEAR AND TURBULENCE — for every landing and departure at Helsinki-Vantaa (research)

What matters to an aircraft near the ground is a sudden change of the **headwind along its own path** — and that is exactly what each aircraft reports: its true airspeed minus its ground speed. The station follows it about **every 0.5 s** for every aircraft on final and on the climb-out (the airspeed from Mode S, the ground speed from ADS-B), finds headwind **ramps** as the Hong Kong Observatory does in its lidar scans, and compares each 200-ft band with the flights of the previous 45 minutes. Every flight gets a stored **windshear report**, and the turbulence (**EDR**, the ICAO measure) of its approach or climb-out from the same replies.

**ADV** advisory ≥ 10 kt **WSA** alert ≥ 15 kt **MBA** ≥ 30 kt loss

`15A WSA 15K+ 2MF`

*"Runway one five arrival, windshear alert, one five knot gain, two mile final"*

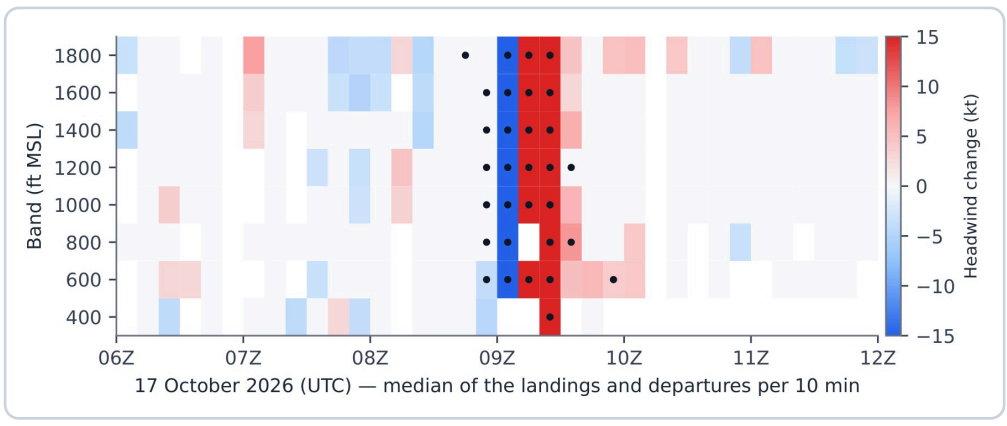
The boundary layer is followed too: the **low-level jet** and the **surface inversion** hour by hour, and every flight's own headwind profile from 200 to 5 000 ft as a column of a time–height section. The results are checked against the METAR windshear groups, the pilots' reports (ARS, WXREP) and the go-arounds — in a **study log** written for aviation meteorologists.

**0.9 %**
of the landings with an advisory, 9–27 Oct 2026

**29 % · 2.4 %**
of the landings in light · moderate turbulence (EDR), 9–29 Oct 2026

**400–1 800 ft**
200-ft bands compared flight to flight

**16 %**
of the night hours with a low-level jet, 29 Sep – 29 Oct 2026

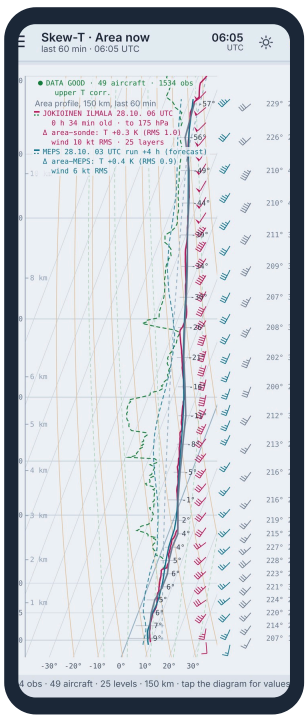



**A front, band by band — 17 Oct 2026.** The headwind change of the landings and departures against the previous 45 min: at 09 UTC a blue streak (less headwind) through every band from 600 to 1 800 ft, then the red reversal after the runway change. A dot: a flight at or beyond its advisory limit.

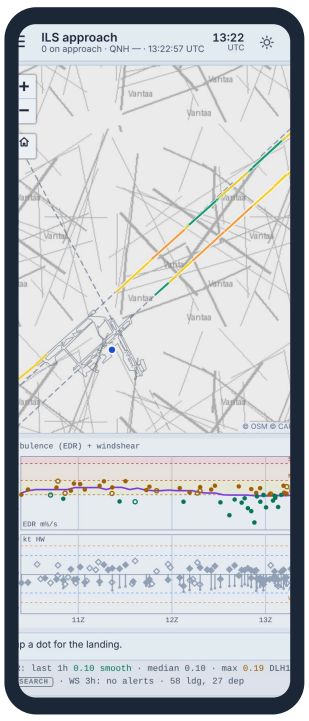


**Wind shear flight by flight, 28 Oct 2026** (Sounding page): each landing a thin column, red where its headwind grows upwards, blue where it falls off — the friction layer by day, from 17 UTC the top of a low-level jet at 2 000–2 600 ft.

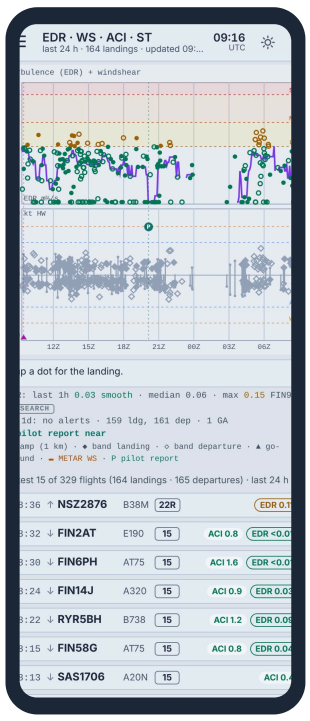
## 5 ON YOUR PHONE — mode-s-wind.oh2gax.org/app: one view at a time, no app store needed



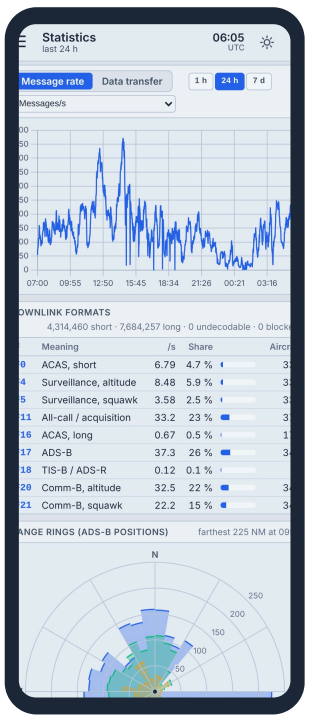

**Skew-T · area now**
49 aircraft of the last hour with the radiosonde and the model



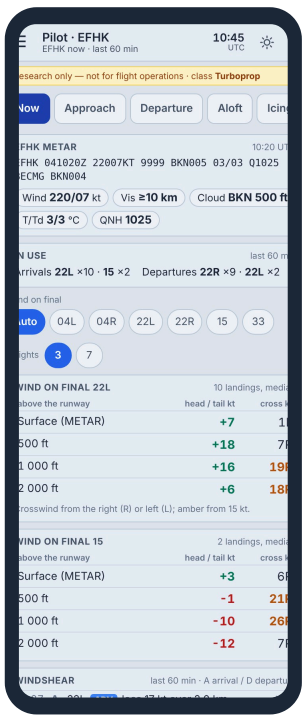

**ILS approach**
the approach map with every flight's turbulence, the profile and strips




**EDR · WS · ACI · ST**
turbulence and windshear of 24 h, the pilot report (P), the flights




**Statistics**
what the receiver hears: rate, formats, range rings




**Pilot · Now**
for pilots (Pilot right): METAR, runways, the wind on final

**mode-s-wind.oh2gax.org/app**
scan with the phone's camera, then *Add to Home Screen*

- **Twelve views** — Skew-T, time–height, ILS, turbulence and windshear, history, statistics, weather, mast, meteogram; GNSS, Pilot (with the right)
- **Tap for values**, portrait or landscape, dark or light
- **Light on data** — about 1–4 MB an hour in the live ILS view

## 6 IN THE BROWSER — the full pages at mode-s-wind.oh2gax.org

**Live Map**
aircraft by meteo source, wind barbs, weather, receiver coverage; an ATC-style radar look

**Sounding**
Skew-T now or any hour, time–height up to 14 days, radiosonde and model checks

**Wind Map**
wind and temperature at a level, streamlines, heights, jet stream, cross-sections

**Weather**
MEPS maps 48 h — rain, gusts, clouds, fog, 2 m temperature — the FMI radar; GFS 16 days, meteograms

**Windshear**
live ILS profile, alerts, go-arounds, approach history, EDR and boundary layer

**GPS**
area-wide GNSS jamming and spoofing seen from the aircraft (Full view users)

**Statistics**
formats, registers, interrogators, range, system health

## See it live

**https://mode-s-wind.oh2gax.org**
phone: **/app** · access with a personal user name — ask the author

**Open data** FMI (soundings, radar) · MET Norway (MEPS) · NOAA (METAR, GFS) · WMM2025 · EGM96
**Open source** github.com/oh2gax/mode_s_wind (GPLv3) · user guides for every page

**Research only.** Passive and uncertified: the alerts describe air already flown through — not for flight operations or air traffic control.

© 2026 Otso Laakso (OH2GAX) · MODE-S Wind, GPLv3 · github.com/oh2gax/mode_s_wind